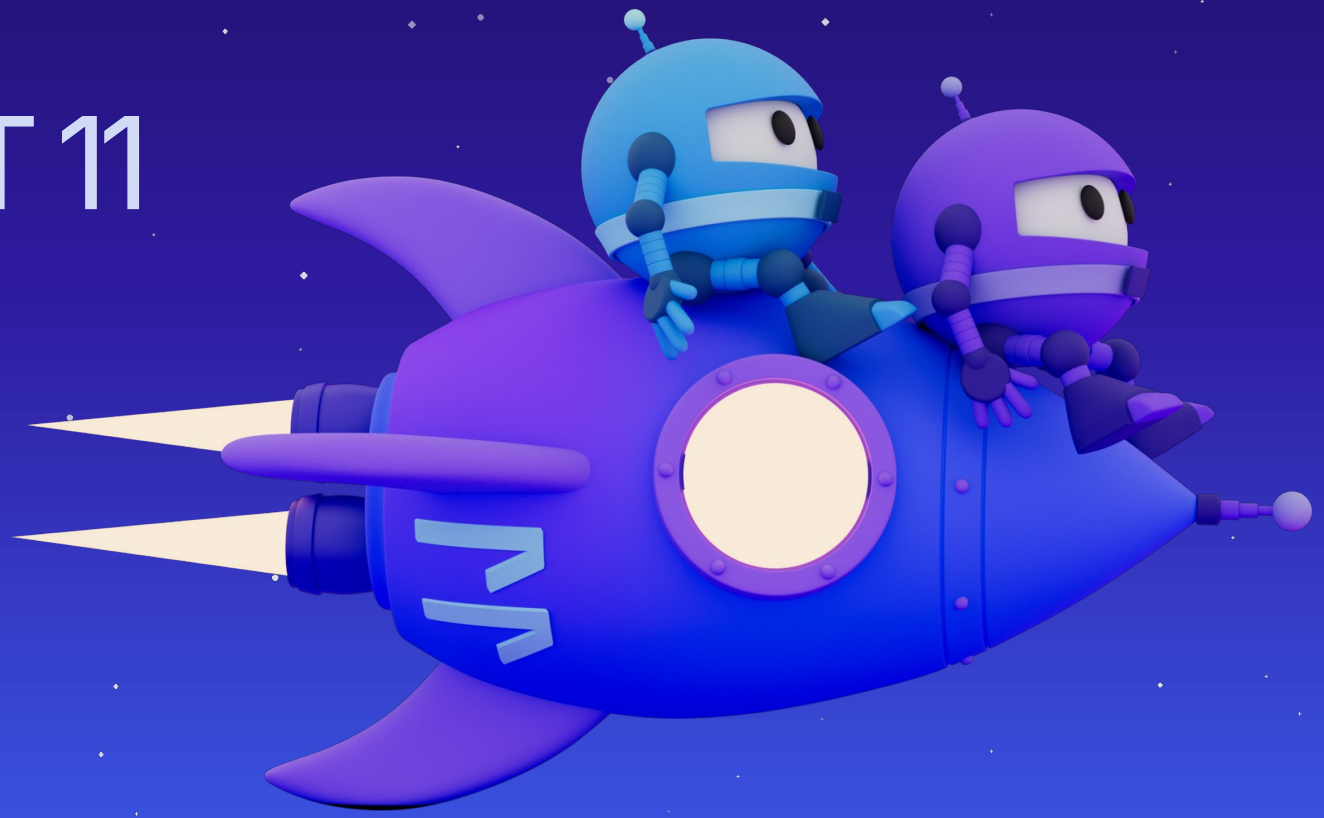
# Infrastructure as (Actual) Code

## with C# 15 and .NET 11

.NET



**Cam Soper**

Systems engineering, Pulumi

# Infrastructure as code, they said

```
resource "azurerm_storage_account" "site" {
  # name, resource group, location, tier...
  account_replication_type = var.env == "prod" ? "ZRS" : "LRS"

  dynamic "blob_properties" {
    for_each = var.env == "prod" ? [1] : []
    content { versioning_enabled = true }
  }
}

resource "azurerm_management_lock" "keep" {
  count = var.env == "prod" ? 1 : 0
  # name, scope, lock level...
}
```

That's not code.
That's config with
extra steps.



# Cam Soper

Systems engineering, Pulumi

13 years at Microsoft: .NET and Azure, co-host of On .NET Live

At Pulumi: built our automated docs QA system. Now: our agentic SDLC

# What “actual code” gets you

## Things C# devs already have

**A compiler**

`error CS8509`

**Types and IntelliSense**

`SkuName.Standard_ZRS`

**Tests**

`dotnet test`

**Packages**

`dotnet add package`

# Pulumi, in one slide

Declare what you want

Pulumi diffs and deploys

Open source, no account:
`pulumi login --local`

```
var group = new ResourceGroup("demo");

var storage = new StorageAccount("site",
    new()
    {
        ResourceGroupName = group.Name,
        Kind = Kind.StorageV2,
        Sku = new SkuArgs
        {
            Name = SkuName.Standard_LRS,
        },
    });
```

# 01 Demo: Hello, cloud



# C# 15: union types

**Exactly one of a fixed set**

**Miss a case: compiler error**

**No default arm needed**

```
public record Dev;
public record Staging;
public record Prod(string OnCall);

public union Stage(Dev, Staging, Prod);

SkuName Redundancy(Stage s) => s switch
{
    Dev     => SkuName.Standard_LRS,
    Staging => SkuName.Standard_LRS,
    Prod    => SkuName.Standard_ZRS,
};
```

# 02 Demo: The compiler is your policy engine



# A rule for my agents. In bold.

From my own CLAUDE.md:

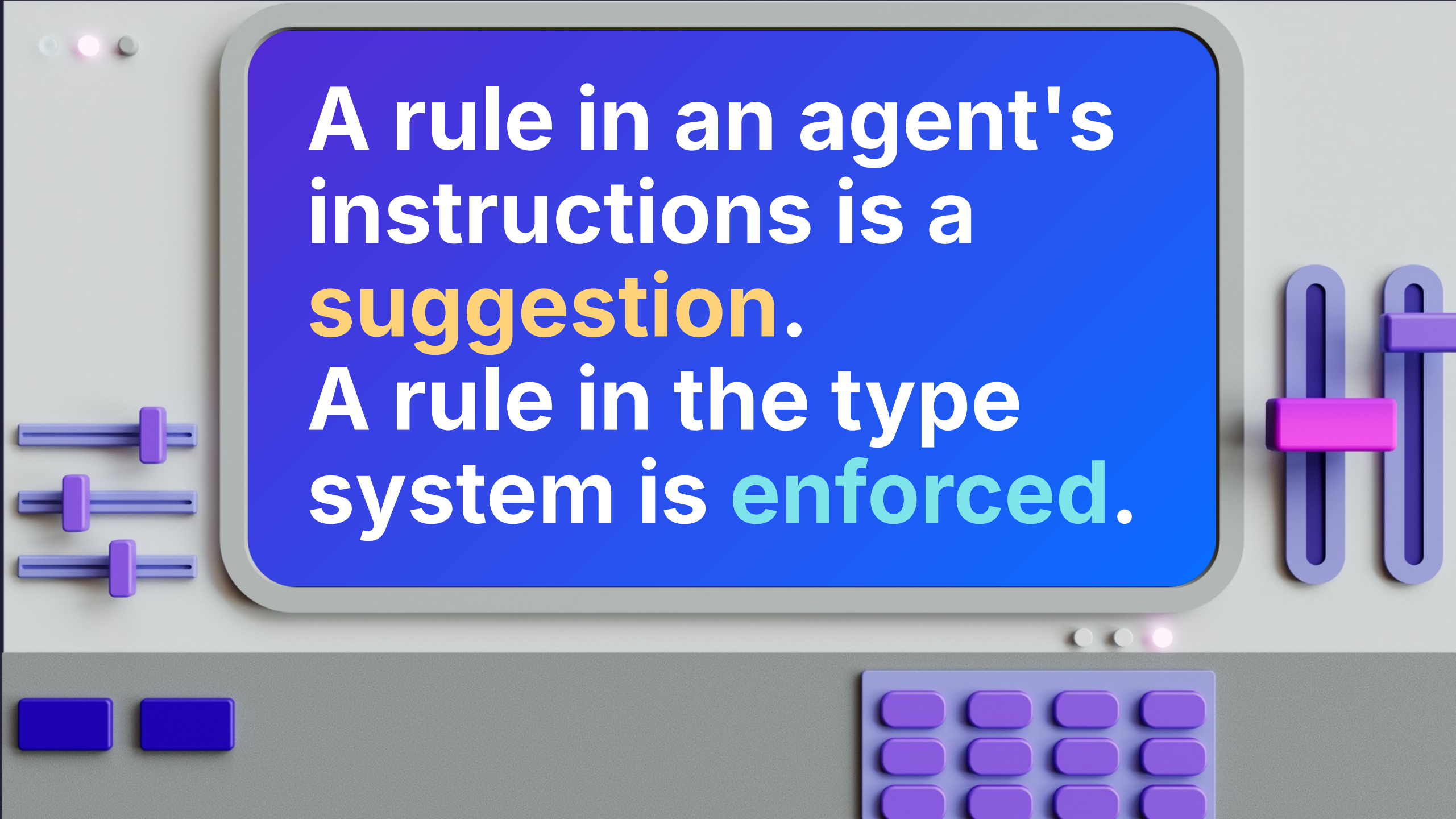
**“Do NOT put Access in front of hades: it self-implements OAuth and would double-prompt.”**

Written by Claude.
Checked in by me. Unread.

```
public union McpAuth(
    SelfOAuth,
    AccessEdgeOnly,
    AccessWithOriginCheck);

record McpService(
    string Name, McpAuth Auth);

var hades = new McpService(
    "hades", new SelfOAuth());
// One service. One posture.
```


A rule in an agent's instructions is a suggestion.
A rule in the type system is enforced.

# 03 Demo: Agent vs. compiler



# Take it home
## Slides, code, and notes

**`demos/`**

Three runnable demos

**`tests/`**

The whole stack vs. a fake cloud. No Azure.

**`bonus/single-file/deploy.cs`**

One file. dotnet run.

**Gotcha**

```
#:property PublishAot=false
```



github.com/CamSoper/infrastructure-as-actual-code

# Thank you

**github.com/CamSoper/infrastructure-as-actual-code**

pulumi.com/docs/iac/languages-sdks/dotnet

“What's new in C# 15” on Microsoft Learn

